

Sakshi

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AI & Machine Learning student with hands-on experience building AI applications, data pipelines, and serverless automation systems. Skilled in Python, SQL, machine learning, and cloud technologies, with a track record of improving data retrieval accuracy, automating workflows, and solving complex engineering challenges.

EDUCATION

BE in Artificial Intelligence and Machine Learning | CGPA – 8.82

Sep 2023 – Present

Cambridge Institute of Technology

EXPERIENCE

System Engineering and Data Analytics Intern

ISRO

Nov 2025 – Present

- Developed geospatial intelligence models using satellite imagery and terrain data to assess urban flood vulnerability in metropolitan regions.
- Managed real-time AWS IoT Core sensor data pipelines, performing data validation, monitoring, and exception handling to ensure reliable data flow.

SKILLS

Languages: Python, SQL, Java, R, JavaScript, C++, HTML/CSS

Data Engineering & Databases: Astra DB (Vector Database), Microsoft SQL Server, DBMS, ETL Data Processing

AI/ML Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy, MLOps, Hugging Face, Keras, SciPy

Developer Tools: Git, Docker, n8n, GCP, AWS IoT Core

Data Visualization: Tableau, Power BI, Matplotlib, Seaborn

Core Concepts: Data Structures & Algorithms, DBMS, Cloud Computing, Computer Vision, CI/CD

PROJECTS

DocuMend– Document QA System | LangChain, Gemini 2.5 Flash, Astra DB, Google Embeddings, Streamlit

- Developed an automated ETL pipeline for processing unstructured data; built and deployed an interactive digital dashboard interface via Streamlit to turn complex semantic contexts into intuitive user insights.
- **20% higher retrieval accuracy** vs. naive chunking, measured via hit-rate evaluation on a held-out query set
- Recursive semantic chunking pipeline reduced irrelevant **context in top-k results by ~30%**

PitchPerfect | Gemini 2.5 Flash, n8n, Astra DB, Airtable, Webhooks, Serverless Automation

- Built a serverless n8n prospect-research pipeline using Gemini 2.5 Flash, Astra DB, web scraping, and API integrations, automating company intelligence gathering and tech-stack analysis to generate personalized outreach drafts, reducing manual **research time by 90%**.
- Designed, monitored, and troubleshoot end-to-end data workflows with custom logging, webhook integrations, and failure diagnostics, ensuring reliable pipeline execution and rapid resolution of synchronization and timeout issues.

Weapon Detector– Real-Time CV System | YOLOv8, PyTorch, OpenCV, Streamlit GitHub

- Fine-tuned YOLOv8N for real-time suspicious object detection in live video; benchmarked YOLOv8n, YOLOv8m, and YOLOv5n using mAP– YOLOv8m achieved best accuracy at acceptable inference latency for edge deployment.

ACCOMPLISHMENTS

• **AMD Slingshot 2026 | Gen AI for Everyone Track | Participant**

Developed a thumbnail analysis tool for content creators using CV and Gen AI; features heatmap visualisation and AI-driven design suggestions to optimise click-through rates.

• **AMD Slingshot Campus days prompt-a-thon | Participant**

Engineered an automated food recommendation engine using Gemini AI and Stitch **under 4 hours**. Integrated Antigravity for rapid prototyping, streamlining the transition from complex user prompts to actionable, data-backed nutritional alternatives.

• **Medal – Awarded by the Cambridge Institute of Technology for academic excellence.**

• **NCC C CERTIFICATE HOLDER | Air wing | 2023 – 2026**

CERTIFICATIONS

System Design – GFG | Prompt Design in Vertex AI– Google Cloud | Gen AI Academy– Hack2Skill | AI Foundation Associate– Oracle Cloud | Data Science for Engineers– NPTEL | DSA with Java– NPTEL